



Lecture 13: Central limit theorem

Yuan Shen

shenyuan_ee@tsinghua.edu.cn

Tsinghua University

May 25, 2021

1. Review
2. Central limit theorem
3. The strong law of large numbers
4. Recitation
5. Summary



1. Review
2. Central limit theorem
3. The strong law of large numbers
4. Recitation
5. Summary



- Some useful inequalities
 - **Markov inequality**

$$\mathbf{P}(X \geq a) \leq \frac{\mathbf{E}[X]}{a}, \quad \text{for all } a > 0.$$

- **Chebyshev's inequality**

$$\mathbf{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

- **Chernoff inequality**

$$\mathbf{P}(X \geq a) \leq e^{-\max_{s \geq 0} (sa - \ln M_X(s))}.$$

- Weak law of large numbers

$$\mathbf{P}(|M_n - \mu| \geq \epsilon) \leq \frac{\text{var}(M_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}$$

- Convergence “in probability”

$$\lim_{n \rightarrow \infty} \mathbf{P}(|Y_n - a| \geq \epsilon) = 0$$

1. Review
2. Central limit theorem
3. The strong law of large numbers
4. Recitation
5. Summary



- $X_1, \dots, X_n$ i.i.d., finite variance σ^2
- “Standardized” $S_n = X_1 + \dots + X_n$:

$$Z_n = \frac{S_n - \mathbf{E}[S_n]}{\sigma_{S_n}} = \frac{S_n - n\mathbf{E}[X_1]}{\sqrt{n}\sigma}$$

$$- \mathbf{E}[Z_n] = 0, \quad \text{var}(Z_n) = 1$$



- $X_1, \dots, X_n$ i.i.d., finite variance σ^2
- “Standardized” $S_n = X_1 + \dots + X_n$:

$$Z_n = \frac{S_n - \mathbf{E}[S_n]}{\sigma S_n} = \frac{S_n - n\mathbf{E}[X_1]}{\sqrt{n}\sigma}$$

- $\mathbf{E}[Z_n] = 0, \quad \text{var}(Z_n) = 1$
- Since the mean and the variance of Z_n remain unchanged as n increases, its distribution neither spreads, nor shrinks to a point.
- The **central limit theorem** is concerned with the **asymptotic shape of the distribution** of Z_n and asserts that it becomes the standard normal distribution.

The Central Limit Theorem

Let $X_1, \dots, X_n$ be a sequence of i.i.d. RVs with common mean μ and finite variance σ^2 ,

$$Z_n = \frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}}.$$

Then, the CDF of Z_n converges to the standard normal CDF

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-x^2/2} dx$$

in the sense that

$$\lim_{n \rightarrow \infty} \mathbf{P}(Z_n \leq z) = \Phi(z),$$

for every z .

- **Usefulness**

- universal and easy to apply; only means, variances matter
- fairly accurate computational shortcut
- justification of normal models

- **What exactly does it say?**

- CDF of Z_n converges to normal CDF
- results for convergence of PDFs or PMFs need more assumptions



- **Usefulness**

- universal and easy to apply; only means, variances matter
- fairly accurate computational shortcut
- justification of normal models

- **What exactly does it say?**

- CDF of Z_n converges to normal CDF
- results for convergence of PDFs or PMFs need more assumptions

- **Normal approximation**

- Treat Z_n as if normal
- hence treat S_n as if normal: $\mathcal{N}(n\mu, n\sigma^2)$

- **Can we use it when n is “moderate”?**

- Yes, but no nice theorems to this effect
- Symmetry and unimodality help

Normal Approximation Based on the Central Limit Theorem

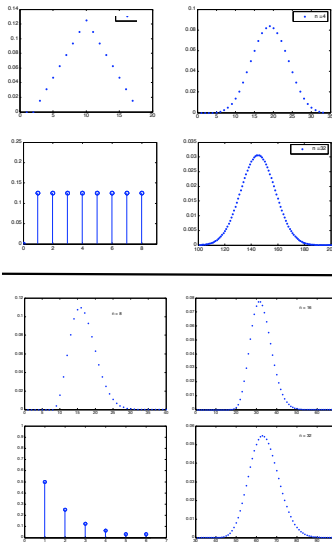
Let $S_n = X_1 + \dots + X_n$, where X_i are i.i.d. RVs with mean μ and variance σ^2 . If n is large, the probability $\mathbf{P}(S_n \leq c)$ can be approximated by treating S_n as if it were normal, according to the following procedure.

- 1 Calculate the mean $n\mu$ and the variance $n\sigma^2$ of S_n .
- 2 Calculate the normalized value $z = (c - n\mu)/(\sigma\sqrt{n})$.
- 3 Use the approximation

$$\mathbf{P}(S_n \leq c) \approx \Phi(z)$$

where $\Phi(z)$ is available from standard normal CDF tables.

Central limit theorem



- Let $X_1, X_2, \dots$ be a sequence of independent identically distributed zero-mean RVs with common variance σ^2 and associated transform $M_X(s)$.
- Let

$$Z_n = \frac{X_1 + \dots + X_n}{\sigma\sqrt{n}},$$

we then have

$$\begin{aligned} M_{Z_n}(s) &= \mathbf{E}[e^{sZ_n}] \\ &= \mathbf{E}\left[\exp\left(\frac{s}{\sigma\sqrt{n}} \sum_{i=1}^n X_i\right)\right] \\ &= \prod_{i=1}^n \mathbf{E}\left[e^{sX_i/(\sigma\sqrt{n})}\right] \\ &= \left(M_X\left(\frac{s}{\sigma\sqrt{n}}\right)\right)^n. \end{aligned}$$

- Suppose that the transform $M_X(s)$ has a second order Taylor series expansion around $s = 0$,

$$M_X(s) = a + bs + cs^2 + o(s^2).$$



- Suppose that the transform $M_X(s)$ has a second order Taylor series expansion around $s = 0$,

$$M_X(s) = a + bs + cs^2 + o(s^2).$$

- Using the moment generating properties of the transform, we have

$$a = M_X(0) = 1, \quad b = \left. \frac{d}{ds} M_X(s) \right|_{s=0} = \mathbf{E}[X] = 0,$$

and

$$c = \frac{1}{2} \cdot \left. \frac{d^2}{ds^2} M_X(s) \right|_{s=0} = \frac{\mathbf{E}[X^2]}{2} = \frac{\sigma^2}{2}.$$

- Suppose that the transform $M_X(s)$ has a second order Taylor series expansion around $s = 0$,

$$M_X(s) = a + bs + cs^2 + o(s^2).$$

- Using the moment generating properties of the transform, we have

$$a = M_X(0) = 1, \quad b = \left. \frac{d}{ds} M_X(s) \right|_{s=0} = \mathbf{E}[X] = 0,$$

and

$$c = \frac{1}{2} \cdot \left. \frac{d^2}{ds^2} M_X(s) \right|_{s=0} = \frac{\mathbf{E}[X^2]}{2} = \frac{\sigma^2}{2}.$$

- Then we have

$$M_{Z_n}(s) = \left(M_X\left(\frac{s}{\sigma\sqrt{n}}\right) \right)^n = \left(a + \frac{bs}{\sigma\sqrt{n}} + \frac{cs^2}{\sigma^2 n} + o\left(\frac{s^2}{\sigma^2 n}\right) \right)^n.$$

- Combining the above parts, it follows that

$$M_{Z_n}(s) = \left(1 + \frac{s^2}{2n} + o\left(\frac{s^2}{\sigma^2 n}\right)\right)^n.$$

- Take the limit as $n \rightarrow \infty$, and use the identity

$$\lim_{n \rightarrow \infty} \left(1 + \frac{c}{n}\right)^n = e^c,$$

to obtain

$$\lim_{n \rightarrow \infty} M_{Z_n}(s) = e^{s^2/2}.$$

- The CDF of Z_n converges to the CDF of a standard normal as the transform $M_{Z_n}(s)$ converges to $e^{s^2/2}$.

- f : the fraction of the entire voter population
- i th (randomly selected) person polled:

$$X_i = \begin{cases} 1, & \text{if yes,} \\ 0, & \text{if no.} \end{cases}$$

- $M_n = (X_1 + \cdots + X_n)/n$
- Suppose we want:

$$\mathbf{P}(|M_n - f| \geq .01) \leq .05$$

- Event of interest: $|M_n - f| \geq .01$

$$\left| \frac{X_1 + \cdots + X_n - nf}{n} \right| \geq .01$$

$$\left| \frac{X_1 + \cdots + X_n - nf}{\sqrt{n}\sigma} \right| \geq \frac{.01\sqrt{n}}{\sigma}$$

$$\begin{aligned} \mathbf{P}(|M_n - f| \geq .01) &\approx \mathbf{P}(|Z| \geq .01\sqrt{n}/\sigma) \\ &\leq \mathbf{P}(|Z| \geq .02\sqrt{n}) \\ &= 2(1 - \Phi(.02\sqrt{n})) \implies n \geq 9604 \end{aligned}$$

- Event of interest: $|M_n - f| \geq .01$

$$\left| \frac{X_1 + \cdots + X_n - nf}{n} \right| \geq .01$$

$$\left| \frac{X_1 + \cdots + X_n - nf}{\sqrt{n}\sigma} \right| \geq \frac{.01\sqrt{n}}{\sigma}$$

$$\begin{aligned} \mathbf{P}(|M_n - f| \geq .01) &\approx \mathbf{P}(|Z| \geq .01\sqrt{n}/\sigma) \\ &\leq \mathbf{P}(|Z| \geq .02\sqrt{n}) \\ &= 2(1 - \Phi(.02\sqrt{n})) \implies n \geq 9604 \end{aligned}$$

- **Comparison:** Use Chebyshev's inequality: If $n = 50,000$, then $\mathbf{P}(|M_n - f| \geq .01) \leq .05$ (conservative)

- A binomial RV S_n with parameters n and p can be viewed as the sum of n independent Bernoulli RVs $X_1, \dots, X_n$ with parameter p ,

$$S_n = X_1 + \dots + X_n.$$

- Recall that

$$\mu = \mathbf{E}[X_i] = p, \quad \sigma = \sqrt{\text{var}(X_i)} = \sqrt{p(1-p)},$$

- A binomial RV S_n with parameters n and p can be viewed as the sum of n independent Bernoulli RVs $X_1, \dots, X_n$ with parameter p ,

$$S_n = X_1 + \dots + X_n.$$

- Recall that

$$\mu = \mathbf{E}[X_i] = p, \quad \sigma = \sqrt{\text{var}(X_i)} = \sqrt{p(1-p)},$$

- Using the approximation suggested by the central limit theorem,

$$k \leq S_n \leq l \Leftrightarrow \frac{k - np}{\sqrt{np(1-p)}} \leq \frac{S_n - np}{\sqrt{np(1-p)}} \leq \frac{l - np}{\sqrt{np(1-p)}}$$

- By the central limit theorem, $(S_n - np)/\sqrt{np(1-p)}$ approximately follows a standard normal distribution,

$$\begin{aligned}\mathbf{P}(k \leq S_n \leq l) &= \mathbf{P}\left(\frac{k - np}{\sqrt{np(1-p)}} \leq \frac{S_n - np}{\sqrt{np(1-p)}} \leq \frac{l - np}{\sqrt{np(1-p)}}\right) \\ &\approx \Phi\left(\frac{l - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right).\end{aligned}$$

- By the central limit theorem, $(S_n - np)/\sqrt{np(1-p)}$ approximately follows a standard normal distribution,

$$\begin{aligned}\mathbf{P}(k \leq S_n \leq l) &= \mathbf{P}\left(\frac{k - np}{\sqrt{np(1-p)}} \leq \frac{S_n - np}{\sqrt{np(1-p)}} \leq \frac{l - np}{\sqrt{np(1-p)}}\right) \\ &\approx \Phi\left(\frac{l - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right).\end{aligned}$$

De Moivre-Laplace Approximation to the Binomial

If S_n is a binomial RV with parameters n and p , with n large and k, l nonnegative integers, then

$$\mathbf{P}(k \leq S_n \leq l) \approx \Phi\left(\frac{l - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right).$$

- Given $n = 36$ and $p = 0.5$, find $\mathbf{P}(S_n \leq 21)$.
- Let S_n be a binomial RV with parameters $n = 36$ and $p = 0.5$.
An exact calculation yields

$$\mathbf{P}(S_n \leq 21) = \sum_{k=0}^{21} \binom{36}{k} (0.5)^{36} = 0.8785.$$

- The central limit theorem approximation yields

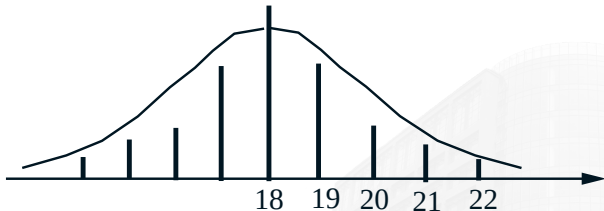
$$\mathbf{P}(S_n \leq 21) \approx \Phi\left(\frac{21 - np}{\sqrt{np(1-p)}}\right) = \Phi\left(\frac{21 - 18}{3}\right) = 0.8413.$$

- Using the 1/2 correction for binomial approximation,

$$\mathbf{P}(S_n \leq 21) \approx \Phi\left(\frac{21.5 - np}{\sqrt{np(1-p)}}\right) = \Phi\left(\frac{21.5 - 18}{3}\right) = 0.879.$$



- $\mathbf{P}(S_n \leq 21) = \mathbf{P}(S_n < 22)$, because S_n is integer
- Compromise: consider $\mathbf{P}(S_n \leq 21.5)$



- When the $1/2$ correction is used, CLT can also approximate the binomial pmf (not just the binomial CDF)

Example:

- We have

$$\mathbf{P}(S_n = 19) = \mathbf{P}(18.5 \leq S_n \leq 19.5)$$

$$18.5 \leq S_n \leq 19.5 \iff$$

$$\frac{18.5 - 18}{3} \leq \frac{S_n - 18}{3} \leq \frac{19.5 - 18}{3} \iff$$

$$0.17 \leq Z_n \leq 0.5$$

- Thus,

$$\begin{aligned}\mathbf{P}(S_n = 19) &\approx \mathbf{P}(0.17 \leq Z \leq 0.5) \\ &= \mathbf{P}(Z \leq 0.5) - \mathbf{P}(Z \leq 0.17) \\ &= 0.6915 - 0.5675 \\ &= 0.124\end{aligned}$$

- **Exact answer:**

$$\binom{36}{19} \left(\frac{1}{2}\right)^{36} = 0.1251$$

1. Review
2. Central limit theorem
3. The strong law of large numbers
4. Recitation
5. Summary



The Strong Law of Large Numbers

Let $X_1, X_2, \dots$ be a sequence of independent identically distributed RVs with mean μ . Then

$$M_n = (X_1 + X_2 + \dots + X_n)/n$$

converges to μ , **with probability 1**, in the sense that

$$\mathbf{P}\left(\lim_{n \rightarrow \infty} \frac{X_1 + \dots + X_n}{n} = \mu\right) = 1.$$

The strong law of large numbers

- Consider the sample space as a set of infinite sequences $(x_1, x_2, \dots)$ of real numbers. The set A consists of those sequences whose long-term average is μ ,

$$(x_1, x_2, \dots) \in A \quad \Leftrightarrow \quad \lim_{n \rightarrow \infty} \frac{x_1 + \dots + x_n}{n} = \mu.$$

- Consider the sample space as a set of infinite sequences $(x_1, x_2, \dots)$ of real numbers. The set A consists of those sequences whose long-term average is μ ,

$$(x_1, x_2, \dots) \in A \quad \Leftrightarrow \quad \lim_{n \rightarrow \infty} \frac{x_1 + \dots + x_n}{n} = \mu.$$

- All the probability is concentrated on this particular subset of the sample space.
- The collection of outcomes that do not belong to A has zero probability.

- The weak law states that the probability

$$\mathbf{P}(|M_n - \mu| \geq \epsilon)$$

of a significant deviation of M_n from μ goes zero as $n \rightarrow \infty$.

- Once in a while, M_n deviates significantly from μ .
- The strong law states that M_n converges to μ with probability 1.
 - For any given $\epsilon > 0$, the probability that the difference $|M_n - \mu|$ will exceed ϵ an infinite number of times is equal to zero.

- Let $X_1, X_2, \dots$ be a sequence of independent RVs that are uniformly distributed in $[0, 1]$, and $Y_n = \min\{X_1, \dots, X_n\}$. Prove that Y_n converges to 0 with probability 1.

- Let $X_1, X_2, \dots$ be a sequence of independent RVs that are uniformly distributed in $[0, 1]$, and $Y_n = \min\{X_1, \dots, X_n\}$. Prove that Y_n converges to 0 with probability 1.

Solution:

- The sequence Y_n is non-increasing and is bounded below by zero, so it must have a limit, which we denote by Y .
- Fix some $\epsilon > 0$, we have $Y \geq \epsilon$ if and only if $X_i \geq \epsilon$ for all i ,

$$\mathbf{P}(Y \geq \epsilon) \leq \mathbf{P}(X_1 \geq \epsilon, \dots, X_n \geq \epsilon) = (1 - \epsilon)^n.$$

- Since it is true for all n , we have

$$\mathbf{P}(Y \geq \epsilon) \leq \lim_{n \rightarrow \infty} (1 - \epsilon)^n = 0.$$

1. Review
2. Central limit theorem
3. The strong law of large numbers
- 4. Recitation**
5. Summary



1. Consider the estimation of the probability p that a bulb produced by the factory is defectless. You are told to assume that all light bulbs have the same probability of having a defect, and that defects in different light bulbs are independent.
 - (a) Suppose that you test n randomly picked bulbs, what is a good estimate Z_n for p , such that Z_n converges to p in probability?
 - (b) If you test 50 light bulbs, what is the probability that your estimate is in the range $p \pm 0.1$?
 - (c) The management asks that your estimate falls in the range $p \pm 0.1$ with probability 0.95. How many light bulbs do you need to test to meet this specification?

2. Random variable X is uniformly distributed between -1.0 and 1.0 . Let $X_1, X_2, \dots$, be independent identically distributed random variables with the same distribution as X . Determine which, if any, of the following sequences (all with $i = 1, 2, \dots$) are convergent in probability.

(a) X_i

(b) $Y_i = \frac{X_i}{i}$

(c) $Z_i = (X_i)^i$



1. Review
2. Central limit theorem
3. The strong law of large numbers
4. Recitation
5. Summary



- Central limit theorem

$$Z_n = \frac{S_n - \mathbf{E}[S_n]}{\sigma_{S_n}} = \frac{S_n - n\mathbf{E}[X]}{\sqrt{n}\sigma}$$

$$\mathbf{P}(Z_n \leq c) \rightarrow \mathbf{P}(Z \leq c), \quad \text{for all } c$$

- The strong law of large numbers

$$\mathbf{P}\left(\lim_{n \rightarrow \infty} \frac{X_1 + \cdots + X_n}{n} = \mu\right) = 1$$

- Readings: Section 5.4-5.5
- Assignment: Problem Set 13
- Next lecture: Bernoulli process; Poisson process